Beyond Real-Valued MRI Features: Complex Broad Learning for Early Alzheimer's Disease Diagnosis

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Clinical Motivation

- Early diagnosis of Alzheimer's disease (AD) enables timely intervention, treatment planning, and improved management of cognitive decline.
- Real-valued MRI representations may not capture coupled interactions between original and complementary latent features.
- Limited clinical samples increase overfitting risk in high-capacity models.
- Resource-constrained clinical settings require expressive features with efficient analytical training.

Main Contributions

- We introduce BLS-X, the first broad learning system designed to operate directly in a complex-valued feature space for early AD diagnosis.
- Two real-to-complex strategies are developed: a natural mapping (BLS-X-N) and an autoencoder-based latent encoding (BLS-X-A).
- Complex-valued feature and enhancement nodes jointly process the real and imaginary components.
- Sparsified randomized mappings reduce overfitting while retaining fast pseudoinverse-based output learning.
- Evaluation covers three ADNI tasks and seven UCI/KEEL benchmark datasets.

MRI Data and Preprocessing

Task	Subjects	Representation	Features
CN vs. MCI	626	VolBM	91
SCD vs. CN	222	Cortical thickness	68
SCD vs. CN	222	White matter	273

Experimental Protocol

- Each dataset is split into 70% training and 30% testing.
- MRI baselines: RVFL, ELM, IFRVFL, NF-RVFL, BLS, NF-BLS, and IF-BLS.
- Generalization baselines: BLS, H-ELM, F-BLS, IF-BLS, GBTSVM, GEIB, and KRPBLS.
- Evaluation uses accuracy and precision for MRI diagnosis, and accuracy, F-Score, and average rank for generalization.

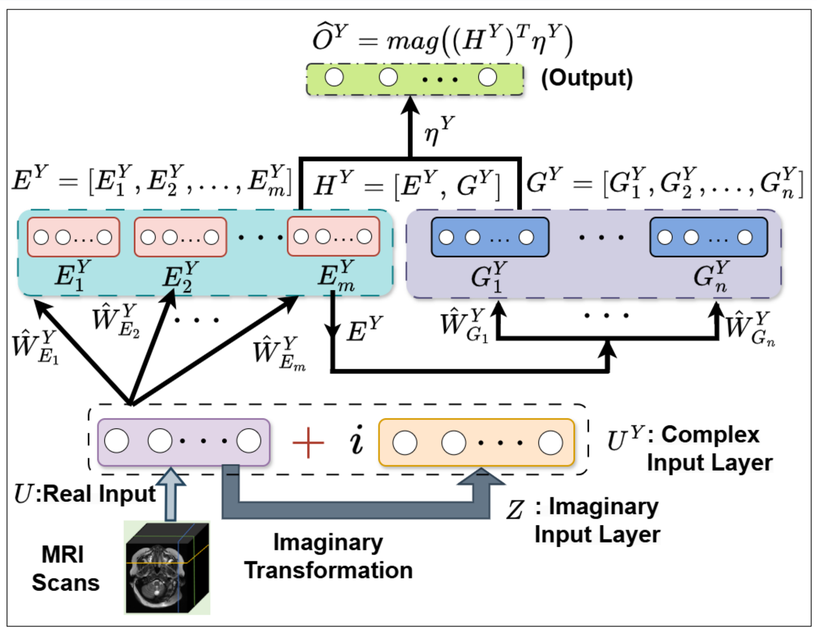
Evaluation Breadth

- The seven UCI/KEEL datasets contain 215–8124 samples, 5–41 features, and imbalance ratios from 0.0455 to 0.9306.
- Hyperparameters are selected using five-fold cross-validation, with $C \in \{10^{-5}, 10^{-4}, \ldots, 10^{5}\}$.
- The benchmarks span biological, medical, ecological, aerospace, and pattern-recognition applications.

Why Complex-Valued Transformation?

- In $U^Y = U + iZ$, the real part retains the original MRI features and the imaginary part contains either zeros or an autoencoder-derived latent representation.
- Complex arithmetic couples both components throughout the feature and enhancement mappings, helping capture subtle AD-related structural variations more richly than real-valued concatenation.

Proposed BLS-X Architecture



Real-to-Complex MRI Representation

Given $U \in \mathbb{R}^{M\times P}$ and $O \in \mathbb{R}^{M\times Q}$, BLS-X forms a complex-valued input U^Y using one of two mappings.

Natural transformation

$$Z = \mathbf{0}_{M\times P},\ U^Y = U + iZ.$$

Autoencoder-based transformation

$$\text{Encoder: } \hat{Z} = \theta(UW_P),$$
$$\text{Decoder: } \hat{U} = \theta(\hat{Z}D_P).$$

$$\text{Optimization: } D_P^* = \arg\min_{D_P} \left[\frac{C}{2}\|U - \hat{Z}D_P\|_F^2 + \frac{1}{2}\|D_P\|_F^2\right],$$

$$\text{Solution: } D_P^* = \begin{cases} \hat{Z}^T\left(\hat{Z}\hat{Z}^T + \frac{1}{C}I\right)^{-1}U, & M < P, \\ \left(\hat{Z}^T\hat{Z} + \frac{1}{C}I\right)^{-1}\hat{Z}^T U, & M \geq P. \end{cases}$$

$$\text{Imaginary Part: } Z = \theta\left(U\left[\delta D_P^* + (1-\delta)(D_P^*)^T\right]\right), \quad \delta \in \{0,1\}.$$

$$\text{Complex-valued Representation: } U^Y = U + iZ.$$

Complex Feature and Enhancement Nodes

$$\text{Complex Weights: } W_{E_j}^Y = W_{E_j}^{\text{real}} + iW_{E_j}^{\text{imag}},\ B_{E_j}^Y = B_{E_j}^{\text{real}} + iB_{E_j}^{\text{imag}}.$$

$$\text{Complex Weights after Sparsity } \alpha\text{: } \hat{W}_{E_j}^Y \text{ and } \hat{B}_{E_j}^Y.$$

$$\text{Enhancement Layer: } E_j^Y = \gamma^Y\left(U^Y\hat{W}_{E_j}^Y + \hat{B}_{E_j}^Y\right),$$

$$\text{Complex Activation Function: } \gamma^Y(z) = \gamma(\text{Re}\, z) + i\,\gamma(\text{Im}\, z),$$

$$\text{Enhancement Group: } E^Y = [E_1^Y, E_2^Y, \ldots, E_m^Y].$$

$$\text{Feature Layer: } G_l^Y = \gamma^Y\left(E^Y\hat{W}_{G_l}^Y + B_{G_l}^Y\right),$$

$$\text{Feature Group: } G^Y = [G_1^Y, G_2^Y, \ldots, G_n^Y],$$

$$\text{Enhanced Group: } H^Y = [E^Y,\ G^Y] \in \mathbb{C}^{M\times(mx+ns)}.$$

Analytical Output Learning

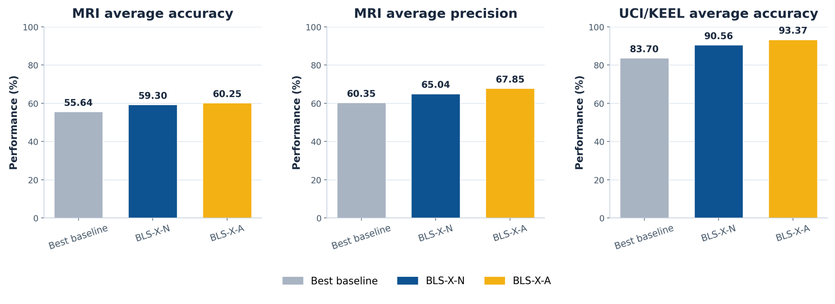
$$\text{Predicted Output: } \widehat{O}^Y = \text{mag}(H^Y\eta^Y),$$

$$\text{Where: } \text{mag}(z) = \sqrt{[\text{Re}(z)]^2 + [\text{Im}(z)]^2}.$$

$$\text{Optimization: } \eta_{\min}^Y = \arg\min_{\eta^Y}\left[\frac{C}{2}\|H^Y\eta^Y - O\|_F^2 + \frac{1}{2}\|\eta^Y\|_F^2\right].$$

$$\text{Solution: } \eta_{\min}^Y = \begin{cases} (H^Y)^T\left[H^Y(H^Y)^T + \frac{1}{C}I\right]^{-1}O, & M < mx + ns, \\ \left[(H^Y)^T H^Y + \frac{1}{C}I\right]^{-1}(H^Y)^T O, & M \geq mx + ns. \end{cases}$$

Performance Overview



Best baseline denotes NF-BLS for the MRI averages and GEIB for the UCI/KEEL average accuracy.

Early-Diagnosis Results (%)

Model	Accuracy: CN vs MCI	SCD (CT)	SCD (WM)	Avg.	Precision: CN vs MCI	SCD (CT)	SCD (WM)	Avg.
RVFL	61.4973	48.4848	48.4848	52.8223	67.1533	43.7500	41.9355	50.9463
ELM	60.9626	46.9697	40.9091	49.6138	68.9394	44.1176	32.5000	48.5190
IFRVFL	49.1979	45.4545	60.6061	51.7528	55.5556	42.3077	55.8824	51.2486
NF-RVFL	63.6364	48.4848	46.9697	53.0303	71.8182	50.0000	40.4255	54.0812
BLS	61.4973	42.4242	54.5455	52.8223	67.4797	36.6667	48.2759	50.8074
NF-BLS	59.3583	53.0303	54.5455	55.6447	67.7165	56.6667	56.6667	60.3500
IF-BLS	63.6364	43.9394	48.4848	52.0202	63.6364	45.7143	40.0000	49.7836
BLS-X-N	**65.7754**	**54.5455**	57.5758	59.2989	**75.2066**	58.6207	61.2903	65.0392
BLS-X-A	62.5668	53.0303	**65.1515**	**60.2495**	73.0159	**61.2903**	**69.2308**	**67.8457**

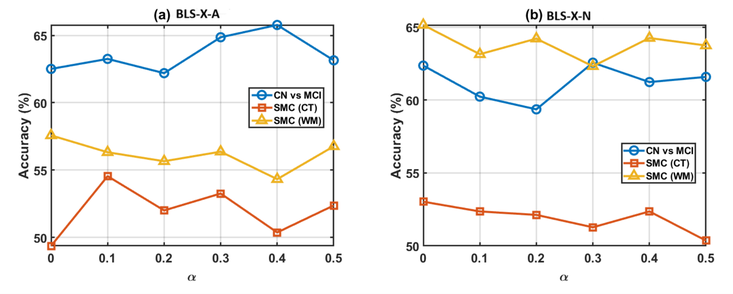
BLS-X-N achieves the best CN vs. MCI accuracy (65.7754%) and precision (75.2066%). BLS-X-A achieves the best overall accuracy (60.2495%) and overall precision (67.8457%).

UCI/KEEL Generalization Results (%)

Model	Accuracy: ECO	LED	MUSH	THY	OOC	SHU	VER	Average
BLS	93.24	73.68	80.69	95.38	56.68	94.20	65.56	79.92
GEIB	84.33	91.67	76.14	96.88	72.55	92.28	72.04	83.70
BLS-X-N	91.78	93.18	99.71	**98.44**	**78.76**	**100.00**	72.04	90.56
BLS-X-A	**97.26**	**96.97**	**99.84**	96.88	**78.76**	**100.00**	**83.87**	**93.37**

BLS-X-A records the best average accuracy (93.37%), F-Score (88.10%), and average rank (1.50). BLS-X-N is second with 90.56%, 79.01%, and 2.25, respectively.

Sensitivity to the Sparsity Parameter



Ablation Study (%)

Table: Importance of Sparsity parameter α: Ablation study of the proposed BLS-X-N and BLS-X-A models .

Models/Dataset	CN vs MCI	SCD (CT)	SCD (WM)	Average
BLS	61.4973	42.4242	54.5455	52.8223
BLS-X-N$_{\alpha=0}$	62.5125	49.3685	57.5758	56.4856
BLS-X-N	65.7754	54.5455	57.5758	59.2989
BLS-X-A$_{\alpha=0}$	62.0656	53.0303	65.1515	60.0825
BLS-X-A	62.5668	53.0303	65.1515	60.2495

Sparsity raises the BLS-X-N average accuracy by 2.81 points and its SCD-CT accuracy by 5.18 points.

Conclusion

- BLS-X extends broad learning to complex-valued MRI representations while retaining efficient analytical learning.
- BLS-X-N gives the best CN vs. MCI accuracy and precision, while BLS-X-A achieves the strongest overall MRI performance.
- BLS-X-A and BLS-X-N achieve the best and second-best average ranks, respectively, on the UCI/KEEL benchmarks.
- Complex-valued mappings provide richer representations without iterative optimization.

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